Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence
Topic: PEAS Framework and Task Environment Analysis

Objectives

The goal of this assignment is to apply the formal frameworks from *Russell & Norvig (Chapter 2)* to a real-world problem. You will demonstrate your ability to specify success metrics, classify the complexity of an AI's operational environment, and represent AI system information in a structured data format.

Task Description

Select **one** of the following modern AI applications:

- **A:** An Automated Warehouse Sorting Robot (e.g., Amazon Robotics).
- **B:** A Personalized News Feed Recommender (e.g., Twitter/X or TikTok algorithm).
- **C:** An Agricultural Monitoring Drone for crop disease detection.
- **D:** A Medical Imaging Assistant for early disease detection (e.g., X-ray triage).
- **E:** A Smart Traffic Signal Controller for reducing congestion in cities.
- **F:** A Voice-Based Virtual Assistant for multilingual customer support.
- **G:** A Fraud Detection System for digital payments and banking transactions.
- **H:** An AI Tutor for personalized student learning recommendations.
- **I:** A Predictive Maintenance System for factory machines and equipment.
- **J:** A Disaster Response Drone for search, mapping, and emergency support.

For your chosen system, complete the following four tasks:

Task 1: PEAS Specification

Define the **PEAS** framework for your agent. Ensure your descriptions are specific (e.g., do not just say "Cameras," specify "High-resolution RGB and Infrared sensors").

Performance Measure: How exactly is success measured? List at least four metrics.

Environment: Where does the agent act? What are the key external factors?

Actuators: What physical or digital tools does the agent use to change its world?

Sensors: What are the inputs the agent uses to perceive its state?

Task 2: Environment Classification

Classify the task environment according to the following dimensions. For each, provide a **one-sentence justification** explaining your choice.

- Fully Observable vs. Partially Observable

- Single-agent vs. Multi-agent
- Deterministic vs. Stochastic
- Episodic vs. Sequential
- Static vs. Dynamic
- Discrete vs. Continuous

Task 3: Critical Analysis

Answer the following questions in short-paragraph format:

1. Which of the six dimensions mentioned in Task 2 poses the **greatest technical challenge** for an AI engineer building this system? Why?
2. Propose one **Utility Function** that would help the agent handle a trade-off (e.g., Speed vs. Safety). How would the agent's behavior change if the weight of this utility was doubled?

Task 4: Structured Data Design (JSON Practice)

Represent your Task 1 and Task 2 outputs in a single valid **JSON** object. This task is for practicing structured data storage and machine-readable AI specifications.

Requirements:

- Include top-level keys: `system_name`, `peas`, `environment_classification`, `utility_function`.
- Under `peas`, include: `performance_measures`, `environment`, `actuators`, `sensors`.
- Store `performance_measures` as an array with at least 4 items.
- Store each environment dimension as a nested object with fields `choice` and `justification`.
- Ensure your JSON is syntactically valid (proper quotes, commas, and brackets).

Suggested JSON template (adapt with your own content):

```
{
    "system_name": "Agricultural Monitoring Drone",
    "peas": {
        "performance_measures": [
            "disease_detection_accuracy",
            "false_alarm_rate",
            "coverage_per_hour",
            "battery_efficiency"
        ],
        "environment": "Large open farmland with variable weather",
        "actuators": ["flight_controller", "sprayer"],
        "sensors": ["RGB_camera", "infrared_camera", "GPS"]
    },
    "environment_classification": {
```

```
        "observability": {
            "choice": "Partially Observable",
            "justification": "Occlusion and weather hide some crop regions"
        },
        "agents": {
            "choice": "Single-agent",
            "justification": "Only one primary decision-making drone"
        }
    },
    "utility_function": "U = 0.6*accuracy - 0.4*energy_cost"
}
```

Submission Guidelines

- Submit your response as a single **PDF** file via email.
- Include your JSON output in a separate section titled **Structured Representation**.
- Format: 12pt font, maximum 3 pages.
- Academic Integrity: All justifications must be your own. Use of AI for formatting is allowed, but the design logic must be original.